6 **(a)** The isotope beryllium-7 $\left({}^{7}_{4}\text{Be}\right)$ is radioactive with a half-life of 53 days. A beryllium-7 nucleus decays by the emission of a γ-ray photon of energy 0.48 MeV.

(i) Calculate the change in mass of a beryllium-7 nucleus when it undergoes γ-decay.

change in mass = ……..…….kg [2]

(ii) Determine the probability of decay of a beryllium-7 nucleus in a time of 1.0 day.

probability = ……..…….. [2]

(iii) Initially, the total mass of undecayed beryllium-7 in a radioactive sample is 5.7×10^{-12} kg.

Calculate the number of undecayed beryllium-7 nuclei in the sample after 120 days.

number = ……..…….. [3]

(iv) Explain why, although the beryllium-7 is radioactive, the number of beryllium-7 nuclei in the sample does not change.

……

……

……

…… [2]

(b) The beryllium-7 nucleus in **(a)** may be considered to be stationary before the emission of the γ-ray photon.

Determine the recoil speed of a beryllium-7 nucleus due to the emission of a γ-ray photon.

speed = ……m s^{-1} [3]

[Total: 12]